

Substrate Skepticism

Objectivity Without Fundamentality

Abstract

Classical skepticism asks whether the world we experience is real. A different question remains even after ordinary objectivity has been granted: does evidence available from within a world determine the physical system by which that world is realized?

This note calls the residual uncertainty *substrate skepticism*. The term is used stipulatively; no claim of terminological priority is intended. The central distinction is

objectivity does not entail fundamentality.

More cautiously, the internally accessible causal organization of a world need not determine its realization relation to a possible host system.

If two realizations preserve every interaction history available to their inhabitants, no internal experiment can distinguish them. The mathematical observation is elementary; its role is to isolate the epistemic gap between knowledge of a world and knowledge of its implementation.

The position is deliberately weaker than full simulation realism. It does not require the claim that simulated objects possess every metaphysical property of non-simulated objects. It requires only that evidence sufficient for ordinary objectivity need not determine realization level.

Whole-brain emulation is not a premise of this separation argument. It serves only as a conditional constructive limit case. Current connectomics and computational neuroscience do not establish conscious brain emulation, but they increasingly connect measured biological structure with executable dynamical models and experimentally testable predictions. If conscious whole-brain emulation were eventually achieved, a traditionally imagined epistemic situation could acquire a constructed instance: an observer could inhabit an objective world while lacking access to the substrate on which that world is realized.

1 Objectivity and realization

Let (W) denote a world accessible to an embedded observer.

Write

$$\text{Obj}(W)$$

for the claim that (W) exhibits ordinary marks of objectivity: persistent objects, stable causal regularities, public accessibility to multiple observers, and reproducible consequences of intervention.

Separately, write

$$\text{Realizes}(H, W)$$

for the claim that a distinct physical system (H) realizes the relevant dynamics of (W) .

One may additionally introduce

$$\text{Fund}(W),$$

meaning that (W) is not itself realized by a distinct underlying system, but the argument does not require a complete metaphysical account of an absolute “bottom level” of reality. The relative realization relation $(\text{Realizes}(H, W))$ is sufficient.

The central claim is therefore modest:

$$\boxed{\text{Obj}(W) \not\Rightarrow \text{Fund}(W).}$$

Equivalently, establishing the objective structure of (W) need not settle whether

$$\exists H; \text{Realizes}(H, W).$$

A successful argument for the objectivity of a world is not automatically an argument for the fundamentality of its realization.

This distinction is closely related to Chalmers’s simulation realism [1]. Chalmers argues that virtual objects may be genuine objects and that inhabiting a simulation need not amount to inhabiting an illusion. The present claim is intentionally weaker. It does not require a full defence of virtual realism. It requires only that ordinary objectivity does not, by itself, determine realization level. Competing views concerning the metaphysics and epistemology of simulation therefore remain compatible with the narrower argument developed here [2].

The residual question is:

$$\boxed{\text{Does evidence available within a world determine what realizes it?}}$$

I use *substrate skepticism* as a convenient label for uncertainty about this question.

2 The formal core

Represent the experimentally accessible history of an embedded observer by

$$T_n = (o_0, a_0, o_1, a_1, \dots, o_n),$$

where (o_i) denotes an observation and (a_i) an intervention.

For a physical realization (R) and an admissible experimental policy (π) , let $\mu_R^{\pi,n}$ denote the probability distribution over such histories.

Two realizations (R_1) and (R_2) are internally observationally equivalent, written $R_1 \equiv_{\text{obs}} R_2$, when

$$\mu_{R_1}^{\pi,n} = \mu_{R_2}^{\pi,n}$$

for every admissible policy (π) and every finite (n) .

The quantification over policies is essential. Agreement along a single actual trajectory would be insufficient. A non-trivial realization must preserve the relevant causal and counterfactual organization of the system, rather than merely permit a retrospective relabelling of physical states.

For deterministic dynamics, one may represent this schematically by a map

$$\iota : S_W \rightarrow S_H$$

satisfying

$$\iota(F_W(s, a)) = F_H^T(\iota(s), \tilde{a})$$

over the relevant states and interventions.

The philosophical literature on computational implementation shows why this requirement matters: unrestricted state mappings risk trivializing the notion of implementation [3]. The present argument does not depend on adopting a permissive implementation theory. Even under a demanding theory of implementation, the underdetermination remains whenever distinct realizations preserve all internally accessible causal and counterfactual structure.

Proposition 1 (Internal indistinguishability). *If*

$$R_1 \equiv_{\text{obs}} R_2,$$

then no decision procedure whose input consists solely of internally available evidence can distinguish (R_1) from (R_2) .

Proof. Any finite internal test can be represented by a measurable decision rule

$$D = f(T_n).$$

For every measurable set (B) of possible outputs,

$$\mathbb{P}_{R_i}(D \in B) = \mu_{R_i}^{\pi,n}(f^{-1}(B)).$$

Since

$$\mu_{R_1}^{\pi,n} = \mu_{R_2}^{\pi,n},$$

we have

$$\mathbb{P}_{R_1}(D \in B) = \mathbb{P}_{R_2}(D \in B).$$

Hence the test has no distinguishing power between the two realizations. $\square$

This proposition is nearly definitional. It is not presented as a substantive mathematical discovery. Its purpose is to expose the structure of the epistemic problem.

Suppose

$$R_F \equiv_{\text{obs}} R_N,$$

while

$$\text{Fund}(R_F) = 1, \quad \text{Fund}(R_N) = 0.$$

Then every internally accessible experimental fact may be identical although the realization status differs.

The substantive separation is therefore

complete internal objectivity $\not\equiv$ knowledge of realization.

Under the stipulated observational equivalence, this is not merely a shortage of data that improved instruments must eventually repair. The missing information is not encoded in the observer's accessible evidence by hypothesis.

3 Why this is not “everything is a dream”

The classical skeptical formulation

“Perhaps everything is a dream”

questions the status of ordinary reality itself.

Substrate skepticism does not.

Suppose a cat remains in a room while an observer looks elsewhere. Other observers can encounter the same cat. Its state evolves according to stable laws. Predictions concerning it can succeed or fail. Experiments can be reproduced.

Then

“the cat exists independently of my present perception”

may be entirely true.

Nothing in substrate skepticism requires denying the cat, the room, the other observers, or the laws connecting them.

The additional question is simply whether

Realizes(H, W)

for some system (H).

The relevant contrast is therefore not

reality versus illusion,

but

objective structure within W versus the realization relation of W .

An embedded scientist may consequently be correct about objects, other observers, and physical laws while remaining ignorant of how the complete system containing those objects is realized.

This is precisely why an argument that successfully rejects a private hallucination need not settle the substrate question.

Substrate skepticism begins after ordinary objectivity has already been conceded.

4 Why neuroscience now matters

The separation argument above is independent of neuroscience.

For most of philosophical history, however, the relevant observer could only be imagined. Contemporary neuroscience has begun to change the empirical background of that thought experiment.

In 2024, Dorkenwald and colleagues reported a wiring diagram of the complete adult female *Drosophila melanogaster* brain containing (139,255) neurons and approximately (54.5) million chemical synapses [5].

Shiu and colleagues subsequently used the adult fly central-brain connectome to construct a brain-wide leaky integrate-and-fire model containing more than (125,000) neurons and roughly (50) million synaptic connections. The model produced predictions about feeding and grooming sensorimotor circuitry, several of which were confronted with optogenetic and behavioural experiments [6].

The relevant methodological sequence is therefore already real:

measured biological structure $\longrightarrow$ executable dynamics $\longrightarrow$ prediction $\longrightarrow$ biological test.

This is not a digital conscious fly. The model is deliberately simplified and establishes neither complete organismal emulation nor consciousness.

At a different scale, the MICrONS project has combined functional recordings from roughly (75,000) neurons in awake mouse visual cortex with an electron-microscopy reconstruction containing more than (200,000) cells and approximately (5×10^8) synapses [7].

The justified empirical conclusion is therefore limited:

increasing amounts of biological causal structure

can be measured, computationally represented, and experimentally interrogated.

It does not follow that

current connectomics $\Rightarrow$ whole-brain emulation.

Nor does it follow that

functional emulation $\Rightarrow$ consciousness.

The importance of these developments for the present note is motivational, not evidential: they make increasingly concrete the possibility of moving from imagined implementations toward constructed ones.

5 The missing equivalence relation

Suppose

$$B_{\text{bio}}$$

is a biological brain and

$$B_{\text{emu}}$$

a proposed computational emulation.

For some functional purpose, one may define an equivalence relation

$$B_{\text{bio}} \sim_F B_{\text{emu}}.$$

The constructive case considered here requires something stronger: an equivalence relation sufficient for preserving conscious experience,

$$B_{\text{bio}} \sim_C B_{\text{emu}}.$$

At present, the content of

$$\sim_C$$

is unknown.

This is more fundamental than asking simply how much resolution a scan must have. The open question is:

Which properties must remain invariant for consciousness to be preserved?

The answer might depend on synaptic connectivity, neuronal dynamics, neuromodulation, cellular physiology, glial interactions, large-scale causal organization, or properties not yet identified.

Current consciousness science supplies no settled bridge principle. A major preregistered adversarial collaboration published in 2025 directly tested predictions of Integrated Information Theory and Global Neuronal Workspace Theory using fMRI, MEG, and intracranial EEG in 256 participants. The results supported some predictions of each framework while substantially challenging important predictions of both [8].

Accordingly, present evidence does not warrant

connectome + executable dynamics $\Rightarrow$ consciousness.

Whole-brain emulation has long been formulated as an engineering programme involving sufficiently detailed brain acquisition, interpretation, computational modelling, and adequate hardware [4]. What level of equivalence would preserve consciousness remains unresolved.

That uncertainty should remain explicit rather than being concealed inside the term *emulation*.

6 The constructive limit case

Whole-brain emulation is not a premise of the separation argument.

Its role is instead:

conceivable countermodel $\longrightarrow$ potentially constructible countermodel.

Introduce one explicitly conditional premise:

C-WBE :

A sufficiently faithful non-biological realization of a particular human brain can instantiate conscious experience while preserving the relevant psychological and causal organization.

Nothing in present connectomics or consciousness science establishes (C-WBE).

Suppose nevertheless that it is true.

Place such an emulated subject in a stable artificial environment (W_S). *The subject may interact with persistence*

Then, by the operational criterion used here,

$$\text{Obj}(W_S) = 1.$$

An external constructor may simultaneously know

$$\text{Realizes}(H, W_S)$$

for some host system (H).

The embedded subject need not be deceived about tables, people, laboratories, or physical laws.

The missing fact concerns only the relation

$$\text{Realizes}(H, W_S).$$

Thus conscious whole-brain emulation, if achieved, would not show that our world is simulated.

It would provide a constructed instance of a narrower proposition:

a conscious observer may inhabit an objective, shared,
law-governed world without thereby knowing what realizes it.

The traditionally imagined epistemic situation would then possess a physical example.

7 Scope

Several stronger conclusions do not follow.

First,

$$\text{Obj}(W) \not\Rightarrow \text{Fund}(W)$$

does not imply

$$\neg \text{Fund}(W).$$

Failure of entailment is not evidence that our world is non-fundamental.

Second,

C-WBE

is a conditional premise, not a consequence of present neuroscience.

Third, the possibility of simulated observers does not imply that our own world is simulated, nor does this note assign any probability to that hypothesis.

Fourth,

$$R_1 \equiv_{\text{obs}} R_2$$

does not imply that all simulations are undetectable. If an implementation produces internally observable artefacts, communication with a host, or any other discriminating evidence, the relevant observational equivalence simply fails.

Finally, nothing here undermines ordinary empirical science. A science of (W) may be entirely successful at identifying the causal organization of (W). What does not follow without an additional bridge principle is that success at describing (W) determines the realization relation of (W).

8 Conclusion

Classical skepticism asks:

Is there an objective world?

Substrate skepticism begins after that question has been answered in the affirmative.

Its question is:

Does everything knowable from within an objective world determine what realizes it?

If distinct realizations can preserve all internally accessible causal and counterfactual structure, then the answer is no.

The conclusion is therefore not that reality is illusory, nor that our world is simulated, nor that consciousness is computational.

It is the narrower separation

$\text{Obj}(W) \not\equiv \text{Fund}(W).$

Current connectomics does not prove that a conscious constructive example is possible. It does, however, make increasingly concrete one part of the path from biological structure to executable dynamical model.

If that path were ever extended to conscious whole-brain emulation, humanity would no longer need merely to imagine an observer who possesses an objective world while lacking epistemic access to its realization.

We could have built one.

The epistemology of a world need not contain the epistemology of its implementation.

References

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